Improving Stratify Domain Adaptation for Scene Text Recognition

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Abstract

Scene text recognition (STR) under domain shift remains a major challenge due to the discrepancy between synthetic training data and real-world images. Stratified Domain Adaptation (StrDA) mitigates this gap via progressive self-training but suffers from costly GAN-based gap estimation, heuristic equal partitioning, and sensitivity to noisy pseudo-labels. We propose two key enhancements: (i) a hierarchical domain gap estimation (HDGE) strategy that divides target samples into five unequal strata from source-like to target-like, allocating more reliable samples to the early stages; and (ii) a domainness–uncertainty driven stratification, inspired by DiaNA, which adaptively clusters target data into four subsets without reliance on GANs. Extensive experiments on standard STR benchmarks demonstrate consistent improvements over StrDA, highlighting the effectiveness of refined stratification and lightweight domain gap estimation for robust domain-adaptive STR. The source code is available at
`https://github.com/quangglammm/StrDA`

Keywords: Optical Character Recognition (OCR), Scene Text Recognition (STR), Unsupervised Domain Adaptation (UDA), Self-Training (ST), Stratified Domain Adaptation (StrDA), Harmonic Domain Gap Estimator (HDGE), Domain Stratification.

1 Introduction

Scene text recognition (STR) is a fundamental task in computer vision with wide applications in document understanding, autonomous driving, and augmented reality. However, STR models often fail to generalize to unseen domains due to large discrepancies between synthetic source data sets and real-world target data. Domain adaptation (DA) has thus become a crucial research direction.

A recent advance in this area, Stratified Domain Adaptation (StrDA) [1], introduces a progressive self-training pipeline that stratifies the target data into multiple subsets based on their domain gap, and gradually adapts the model in a curriculum-like manner. Despite its effectiveness, StrDA still faces important limitations. First, its GAN-based domain gap estimator is computationally costly and unstable. Second, the subset partitioning is heuristic, which may not capture the intrinsic differences among target samples. If the subsets are not chosen appropriately, equal-sized partitioning may lead to overfitting, as increasing the number of subsets reduces the sample size per subset, while STR models typically involve a large number of parameters (Fig. 1). Additionally incurs high computational cost, including time and storage overhead, since extensive hyperparameter tuning for the number of strata requires training multiple models before selecting the best one.

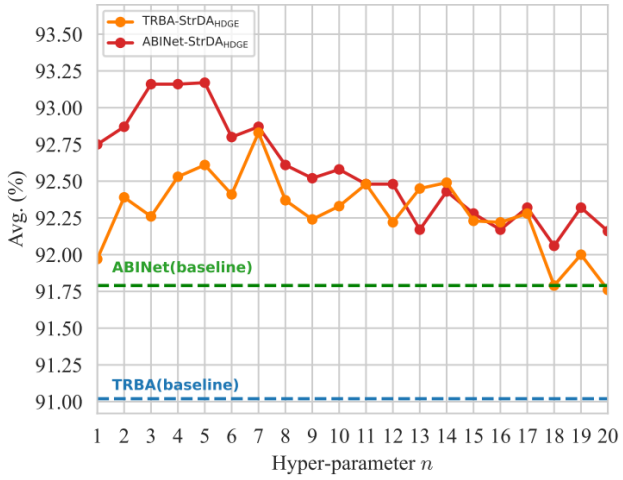

Figure 1: Visualization of subset numbers hyperparameter impact on target domain adaptation of STR model in StrDA.

To address these issues, we propose two complementary strategies to refine the stratification mechanism in StrDA:

- **Strategy 1 (Stage 1):** We retain the Harmonic Domain Gap Estimator (HDGE) from StrDA[1], but instead of dividing the target data into equal-sized subsets, we apply a divide-and-conquer partitioning scheme. Specifically, target samples are stratified into subsets of varying sizes, with larger subsets assigned to samples closer to the source domain and progressively smaller subsets assigned to samples farther away, that is, closer to the target domain. This design enforces a natural curriculum where the model adapts from easier to harder samples, while maintaining a balanced learning pace.

- **Strategy 2 (Stage 1):** Inspired by DiaNA[2], we compute domainness and certainty scores for each predicted character, and use these to group target data into four subsets. This approach captures both domain discrepancy and prediction confidence, leading to a more adaptive

stratification than heuristic partitioning.

In summary, our contributions are two-fold:

1. We propose an enhanced HDGE-based subset partitioning strategy, which creates subsets of decreasing sizes to better reflect the progression from source-like to target-like samples.

2. We introduce a domainness–uncertainty–driven stratification, inspired by DiaNA, to provide a complementary and more adaptive criterion for target-sample clustering.

Extensive experiments demonstrate that these strategies produce consistent improvements over the baseline StrDA, highlighting the importance of refined stratification for robust domain-adaptive scene text recognition.

2 Related Work

2.1 Scene Text Recognition

Scene Text Recognition (STR) has been widely studied as a sequence prediction problem, where the goal is to map visual features into character sequences. Early approaches combined convolutional backbones with recurrent decoders and Connectionist Temporal Classification (CTC) loss[3], achieving strong performance on constrained benchmarks. Attention-based decoders[4,5] later improved recognition flexibility by enabling explicit sequence modeling. More recently, transformer-based models such as ViTSTR[6] and PARSeq[7] have demonstrated superior accuracy and scalability, while vision-language pretraining (e.g., TrOCR[8], Donut[9]) further pushed the limits by leveraging large-scale multimodal data.

Despite these advances, STR models trained purely on synthetic data remain sensitive to domain shifts, particularly when faced with real-world images containing blur, occlusion, or diverse fonts.

2.2 Unsupervised Domain Adaptation for STR

To overcome domain discrepancies without requiring target labels, unsupervised domain adaptation (UDA) has been extensively explored in computer vision. Typical approaches include adversarial domain alignment[10], distribution matching via maximum mean discrepancy (MMD)[11], and contrastive representation learning. In the context of STR, UDA methods usually adopt a self-training pipeline: train on labeled source data, generate pseudo-labels for unlabeled target data, and refine the model using those pseudo-labels. Variants include uncertainty-aware label filtering[12], iterative re-labeling, and curriculum-based adaptation. Baek et al[13] showed that carefully crafted pseudo-labels can yield significant improvements, motivating further refinements in confidence estimation and sample selection.

2.3 Stratified Domain Adaptation

A particularly influential work is Stratified Domain Adaptation (StrDA)[1]. StrDA introduces the Harmonic Domain Gap Estimator (HDGE) to quantify the similarity of each target sample to the source domain, ranks samples accordingly, and progressively incorporates them into training from the most to the least source-like. This curriculum-style adaptation enhances stability and improves performance across multiple STR backbones.

Despite their progress, StrDA for STR remain limited. First, GAN-based gap estimation in StrDA can be computationally costly and unstable. Second, dividing target samples into equal-sized strata is heuristic and may not reflect the intrinsic distribution of target data. These limitations motivate our proposed refinements, which aim to design more principled stratification strategies that balance computational efficiency and adaptation robustness. Recent work such as DiaNA[2] has also explored stratification strategies using domainness and certainty as clustering criteria, offering an alternative to distribution-based gap estimation.

Algorithm 1 Divide-and-Conquer Stratification of Target Data

Require: Sorted index list of target samples $\mathcal{I}$, number of strata n
Ensure: Subsets $\{S_1, S_2, \ldots, S_n\}$
Initialize empty list $\mathcal{S} \leftarrow [\ ]$
Set $\mathcal{J} \leftarrow \mathcal{I}$ // current indices to split
for $i = 1$ to $n - 1$ **do**
  $m \leftarrow \lfloor |\mathcal{J}|/2 \rfloor$
  Append first half $\mathcal{J}[1 : m]$ to $\mathcal{S}$
  $\mathcal{J} \leftarrow \mathcal{J}[m + 1 : end]$ // continue splitting the harder part
end for
Append remaining $\mathcal{J}$ to $\mathcal{S}$ as S_n
return $\mathcal{S} = \{S_1, S_2, \ldots, S_n\}$

3 Methods

We focus on improving Stage 1 of Stratified Domain Adaptation (StrDA) by introducing more principled stratification strategies. In the original StrDA, the target domain is divided into equal-sized subsets according to a domain gap score. However, this equal partitioning neglects the varying reliability of pseudo-labels across subsets and requires manual tuning of the number of subsets, which is costly in terms of computation and storage. Our objective is to design 2 stratification methods that (i) allocate more reliable samples to earlier strata and (ii) adaptively separate target data based on both domain shift and prediction confidence. These improvements lead to more stable progressive adaptation in Stage 2.

3.1 Method 1: Divide-and-Conquer Stratification

We first propose a hierarchical divide-and-conquer strategy that recursively partitions the target dataset into subsets of increasing difficulty. Unlike StrDA's uniform partitioning, this method ensures that earlier strata contain more samples than later ones, reflecting the intuition that high-quality pseudo-labels should dominate the initial training phase.

Given a stratification parameter s, the target dataset $\mathcal{D}_t$ is repeatedly split into two halves along the domain gap axis until s subsets are obtained. At each split, the left child corresponds to easier samples (closer to the source domain) and is fixed as

Type	Method	IIIT5k_3000	SVT_647	IC13_1015	IC13_857	IC15_1811	IC15_2077	SVTP_645	CUTE80_288	AVG
CRNN	Baseline	80.1	77.59	87.78	89.73	63.11	58.11	61.4	61.81	72.43
	StrDA	**84.13**	79.91	**89.36**	**90.32**	66.7	61.68	64.96	68.4	**75.70**
	Method 1	82.73	79.75	88.18	89.38	65.99	61.24	**66.36**	65.97	74.89
	Method 2	83.47	**80.53**	88.57	89.5	**67.2**	**62.16**	65.58	**68.06**	75.61
TRBA	Baseline	85.53	85.78	90.05	91.6	72.45	66.63	73.8	69.44	79.23
	StrDA	89.5	89.49	92.32	**94.17**	**78.69**	**72.56**	**81.71**	78.82	**84.08**
	Method 1	88.6	89.18	**92.71**	93.82	77.14	71.11	80.16	**80.56**	83.20
	Method 2	**89.57**	**89.64**	91.92	93.58	78.02	72.12	79.38	80.21	83.71

Table 1: Word accuracy on multiple scene-text benchmarks. The number of words in each dataset is listed with its name. "Baseline" means models trained only on synthetic data. StrDA is the method proposed in [1]. Method 1 and 2 are our proposed methods. With each column, the best result is shown in bold.

one stratum, while the right child is recursively divided. As a result, the final partition $\{\mathcal{D}_t^{(1)}, \ldots, \mathcal{D}_t^{(s)}\}$ is imbalanced, with $\mathcal{D}_t^{(1)}$ being the largest and $\mathcal{D}_t^{(s)}$ the smallest (Algorithm 1).

This imbalance is intentional: the first stratum provides a large pool of relatively accurate pseudo-labels, reducing the risk of overfitting and enabling the model to generalize better before gradually incorporating more difficult samples.

3.2 Method 2: Domainness - Uncertainty Driven Stratification

To capture richer structure in the target data, we extend StrDA with a DiaNA-inspired mechanism that jointly considers domainness and uncertainty[2]. This method evaluates each target sequence along two axes: (i) similarity to the source domain and (ii) reliability of its pseudo-label.

3.2.1 Domainness

For each source character class k, we compute a centroid c_k as the mean contextual embedding. Given a target token feature $f_{i,t}$, its distance to the closest centroid is measured, and the sample-level domainness D_i is obtained by averaging over tokens. Lower D_i indicates stronger alignment with the source distribution.

3.2.2 Uncertainty

For a target sequence with token-level logits, we compute predictive distributions via softmax and measure entropy at each position. The mean entropy across tokens defines the sequence-level uncertainty U_i. Lower U_i corresponds to more confident predictions.

3.2.3 Joint Partitioning

Each sample is represented as a normalized 2D vector $[\tilde{U}_i, \tilde{D}_i]$. A Gaussian Mixture Model (GMM) with $K = 4$ components is fitted on all samples, producing four semantic subsets:

- *cc (confident–consistent)*: low domainness, low uncertainty.
- *uc (uncertain–consistent)*: high domainness, low uncertainty.
- *ci (confident–inconsistent)*: low domainness, high uncertainty.
- *ui (uncertain–inconsistent)*: high domainness, high uncertainty.

This stratification prioritizes samples in the *cc* subset, which are both reliable and source-aligned, while deferring noisier or more shifted samples (*ui*) to later training stages. By jointly modeling prediction certainty and domain gap, this approach produces more coherent strata than equal partitioning or domainness-only methods.

4 Experiments

Experiments are conducted according to the setup of StrDA [1]. However, due to resource limitation, we create a subset of training and validation datasets to reduce the number of training samples. To conduct a fair comparison, we have fixed the choice of training, validation, and evaluation datasets, and reproduced the result from StrDA [1].

4.1 Dataset

Training data: We used synthetic data (with SynthText (ST) [14] and MJSynth (MJ) [15]) for source domain and unlabeled real data for target domain. Concretely, for real dataset, we use ArT [16], TextOCR [17], and discard their labels. In addition, as in STrDA [1], we exclude vertical text ($height > width$) and images whose width is greater than 25 times the height, also we only keep the English words. In total, we have 1 million labeled synthetic data and 100000 unlabeled real data for training, compare with 16 million labeled synthetic data and 2 million unlabeled real data in StrDA [1].
Validation data: We used synthetic data with a total of 200000 samples from SynthText (ST) [14] and MJSynth (MJ) [15] for validation dataset during training models.
Evaluation data: We used six standard benchmark datasets, including IIIT 5k-word (IIIT) [18], Street View Text (SVT) [19], ICDAR 2013 (IC13) [20], ICDAR 2015 (IC15) [21], SVTPerspective (SVTP) [22], and CUTE80 (CUTE) [23]

4.2 Evaluation Metrics

Following StrDA [1], we present word-level accuracy for each dataset and an average score across all evaluation datasets.

4.3 Implementation Details

To assess the effectiveness of the proposed framework and compare with StrDA [1], we use 2 STR models: CRNN [24] and TRBA [25]. We trained the baseline STR models in a

fully supervised manner using the synthetic dataset (MJ+ST). We adopt the AdamW optimizer (weight decay 0.005). We also use the one-cycle learning rate scheduler with a maximum learning rate of 0.0005. Our training batch size is fixed at 128, and the total number of epoch is 25. Image augmentation is also applied.

For Stage 1 (Domain Stratifying), we create subsets based on 3 methods: Stratified Domain Adaptation as in [1], Divide-and-Conquer Stratification and Domainness - Uncertainty Driven Stratification.

For Stage 2 (Progressive Self-Training), we use the same training setting with supervised based model learning above.

4.4 Results

As illustrated in Table 1, all methods surpass the baseline models across all the benchmark datasets. However, our proposed methods are still left behind in average accuracy compare in the original method as in [1]. Using CRNN base model, the method 2 (Domainness - Uncertainty Driven Stratification) surpass StrDA [1] in half of the benchmark, with average accuracy very close to StrDA. With TRBA base model, the difference become larger, and StrDA is better in most of the benchmark datasets.

5 Conclusions

In this work, we explored two alternative stratification solutions for unsupervised domain adaptation in scene text recognition. While both methods yielded improvements on certain datasets, they did not consistently surpass the average accuracy of the original StrDA framework. Nevertheless, given the limited training data in our experiments, it is noteworthy that Solution 2 was able to closely match the performance of StrDA, suggesting strong potential to outperform the original approach when trained on larger datasets.

Moreover, the DiaNA-inspired method demonstrated the advantage of reducing computational costs by avoiding HDGE, while retaining adaptability to different model architectures.

Future work will focus on training and evaluating these methods on larger benchmark datasets to further assess their effectiveness and scalability in real-world scenarios.

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